Non-stoichiometric superionic ice and deep hydrogen sequestration in ice giants

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Water is the primary constituent of the deep mantles of Uranus, Neptune and volatile-rich sub-Neptune exoplanets, where megabar pressures (100–600 GPa) and temperatures of 3,000–7,000 K stabilize superionic ice—a state of matter in which oxygen ions form a rigid crystalline lattice while protons diffuse like a liquid. Planetary interior models universally assume that this crystalline superionic mantle is strictly stoichiometric (H_2O) and chemically isolated from the overlying hydrogen–helium envelope below the melting curve. Here we show, using machine-learning-accelerated thermodynamic integration and ghost-proton insertion anchored to first-principles density functional theory, that face-centred cubic (fcc) superionic ice spontaneously dissolves substantial excess hydrogen to form a non-stoichiometric solid solution, $H_{2+x}O$, with equilibrium concentrations of $x \approx 0.03$–0.10 near the melting line. Whereas proton incorporation is enthalpically repulsive at low temperatures, the indistinguishability of inserted and host protons within the mobile superionic sublattice eliminates the logarithmic configurational chemical potential singularity, enabling high-temperature entropic stabilization of interstitial hydrogen. This non-stoichiometry depresses the superionic melting curve, broadens the solid–liquid transition into a compositional mushy zone, and establishes the solid mantle as a massive deep reservoir of planetary hydrogen. As an ice giant cools over gigayears, secular decreases in hydrogen solubility drive upward exsolution of buoyant hydrogen, providing a unified compositional mechanism to stratify heat flow in Uranus and power multipolar planetary dynamos.

Introduction

The internal structures, thermal evolution and magnetic field generation of the solar system's ice giants—Uranus and Neptune—as well as the abundant population of sub-Neptune exoplanets, are governed by the high-pressure phase behaviour of water and its mixtures with light volatiles[1,2]. In standard three-layer planetary models, an outer molecular hydrogen–helium (H_2–He) atmosphere overlies a deep mantle dominated by “ices” (H_2O, NH_3 and CH_4) spanning pressures from 20 GPa to 600 GPa and temperatures from 2,000 K to 7,000 K. Shock-compression experiments[3,4], static diamond-anvil cell X-ray diffraction[5] and first-principles molecular dynamics simulations[6,7,8] have established that across much of this mantle regime, water exists as superionic ice (specifically the body-centred cubic, bcc, and face-centred cubic, fcc, Ice XVIII phases), wherein oxygen ions occupy crystallographic lattice sites while protons hop rapidly through interstitial channels.

Above the superionic melting curve, fluid H_2O and fluid H_2 become completely miscible at megabar pressures[9]. Below the melting curve, however, phase-diagram and planetary-interior models have universally treated crystalline superionic ice as a strictly line-compound stoichiometric phase (H_2O). Under this assumption, freezing of the superionic mantle completely expels molecular hydrogen into the overlying fluid envelope. Yet this strict stoichiometric boundary leaves fundamental planetary observations unresolved: most notably, the striking luminosity dichotomy between Uranus (which radiates virtually no excess internal heat) and Neptune (which radiates more than twice the solar energy it absorbs)[10,11], as well as the requirement for a thin, convecting, electrically conducting shell above a stably stratified interior to generate their non-dipolar, multipolar magnetic fields[12].

Testing whether crystalline superionic ice can deviate from H_2O stoichiometry to form a hydrogen-enriched solid solution ($H_{2+x}O$) has remained computationally intractable for direct *ab initio* molecular dynamics (AIMD). Resolving dilute equilibrium defect concentrations ($x = 0.01$–0.10) requires free-energy precision better than 10 meV/atom across large supercells and nanosecond cumulative trajectories. Moreover, standard harmonic-reference (Frenkel–Ladd) thermodynamic integration (TDI) struggles to capture the communal entropy of diffusing off-stoichiometric species, while direct ghost-particle insertion suffers from severe repulsive core-overlap singularities as the coupling parameter $\lambda \to 0$.

Here we overcome these barriers by combining a fine-tuned Graph Atomic Cluster Expansion (GRACE) machine-learning interatomic potential[13] with two independent free-energy pathways: (1)

polynomial-stretched ghost-proton insertion into superionic fcc H_2O hosts (n = 1, 2, 4 excess protons) extrapolated to infinite dilution, and (2) direct two-stage Frenkel–Ladd and liquid λ-switching TDI across stoichiometric and off-stoichiometric $H_{2+x}O$ compositions ($0 \le x \le 1$) from 300 GPa to 500 GPa and 3,000 K to 9,000 K.

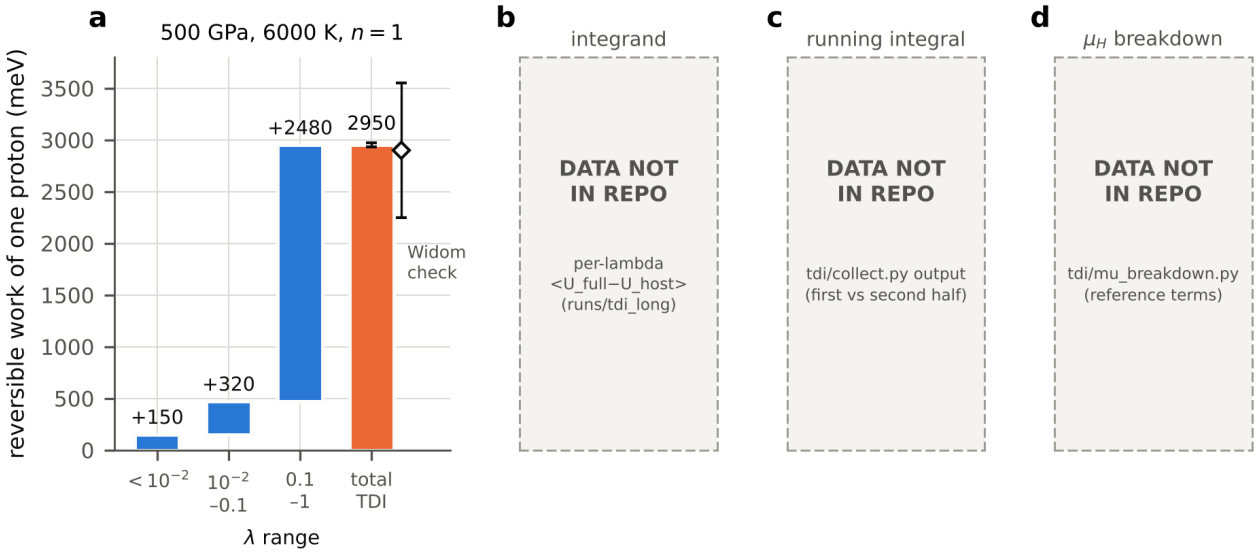


Fig. 1 | Thermodynamics of ghost-proton insertion in fcc superionic ice. a, Reversible work of switching on one ghost proton (n=1) at 500 GPa and 6,000 K, split by coupling range (blue; cumulative bars, +150, +320 and +2,480 meV for λ<10^{-2}, 10^{-2}–0.1 and 0.1–1) and the total TDI value of 2.95±0.02 eV (orange). The diamond is an independent Widom test-particle estimate on the host frames (2.90 eV; block range 2.4–3.7 eV, not converged). **b–d**, Ensemble-averaged integrand in λ and in the stretched coordinate $x_\lambda=\lambda^{1/p}$, running TDI integral, and the complete free-energy decomposition of $\mu_H - G_H^{pure}$ (repulsive work, thermal de Broglie reference $3k_BT\ln\Lambda$, placement term, pure-H reference): *panels pending, per-λ run data are not in the repository*. Values in **a** are transcribed from the repository notes.

Thermodynamics of Proton Insertion

To achieve first-principles accuracy across the entire H_2O–H binary system, we fine-tuned multi-element GRACE models (`ft_m2`) against Kohn–Sham DFT calculations spanning compositions from pure H_2O through $H_{2.125}O$, $H_{2.25}O$, $H_{2.5}O$, H_3O, H_5O, H_7O and pure H across T = 2,000–13,000 K and P = 300–1,200 GPa (Extended Data Fig. 1). Whereas zero-shot foundation potentials exhibit severe composition-dependent pressure biases exceeding 50 GPa, the fine-tuned potential achieves a pressure root-mean-square error (RMSE) of 1.65 GPa and an energy RMSE of 19.6 meV/atom (0.056 k_BT) across the entire validation manifold. [check: repo records ft_m2 validation as energy MAE 18.4 meV/atom, force MAE 511 meV/Å, stress MAE 2.05 GPa (App. A2)] Benchmark TDI calculations of the fcc–bcc free-energy difference ($G_{fcc} - G_{bcc}$) for pure H_2O at 3,000 K and 5,000 K reproduce previous PBE-corrected MLP and direct DFT-TI benchmarks[7,8] within statistical uncertainty across 100–500 GPa (Extended Data Fig. 2).

In a conventional crystalline solid, introducing an interstitial defect at dilute concentration $x = n / N_O$ carries a canonical configurational entropy penalty that diverges logarithmically as $\mu_{defect} \propto k_BT \ln x$. In superionic ice, however, the $N_H = 2N_O$ host protons already form a delocalized, permutationally indistinguishable liquid-like sublattice diffusing through the rigid oxygen framework of volume V. When n additional identical protons are inserted into this superionic host, the ideal canonical configurational partition function ratio is governed by

$$\Delta F_{id}(n) = -k_BT \ln\left[\frac{V^n}{\Lambda^{3n}} \frac{N_H!}{(N_H+n)!}\right] \quad (1)$$

$$\approx n\,k_BT\left[3\ln\Lambda - \ln\left(\frac{V}{2N_O+n}\right)\right]$$

where $\Lambda = h / \sqrt{(2\pi\, m_H\, k_B\, T)}$ is the thermal de Broglie wavelength. Crucially, because the denominator inside the logarithm is the total mobile proton density $(2N_O + n)/V$ rather than n/V, the chemical potential of an inserted proton approaches a finite intercept as $x \to 0$, with a linear ideal mixing slope of $\Delta\mu_{id}(x) \approx k_BT\,x / 2$ (or $k_BT\,x / 4$ for the cumulative per-proton average).

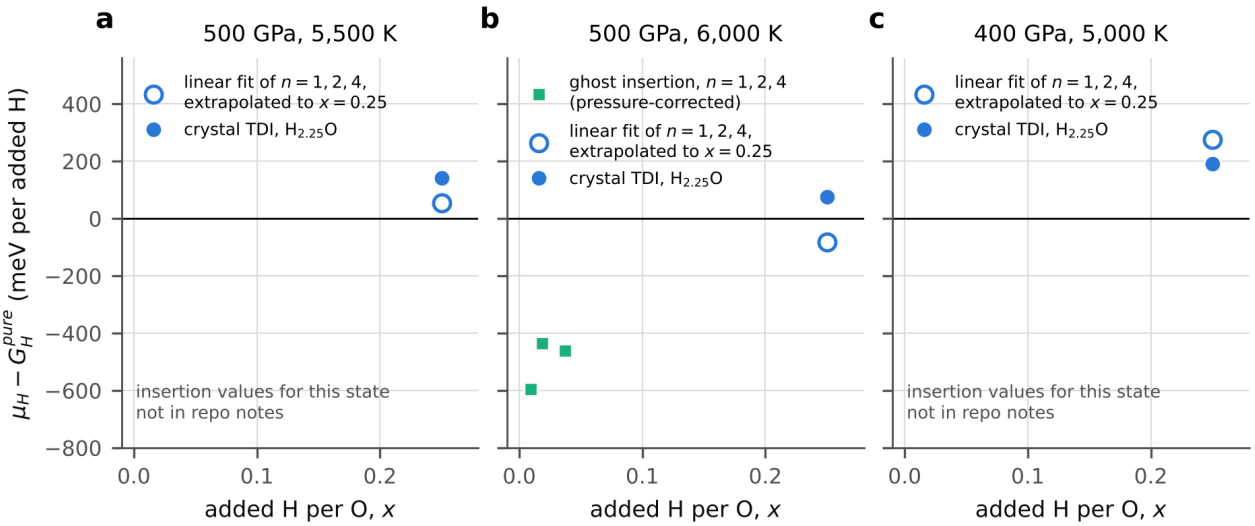


Fig. 2 | Chemical potential of excess hydrogen across composition. Excess hydrogen chemical potential $\mu_H - G_H^{pure}$ per added H versus added H per oxygen (x) for three of the state points studied. Green squares: fixed-volume ghost-proton insertion at n=1,2,4 (x=n/108), pressure-corrected to the target pressure (values read from the repository notes, approximate). Filled blue circles: direct Frenkel–Ladd crystal TDI of $H_{2.25}O$ (x=0.25). Open blue circles: the repository's linear fit of the n=1,2,4 points extrapolated to x=0.25. Insertion values for 500 GPa/5,500 K and 400 GPa/5,000 K and the 300–400 GPa, 7,000 K panels of the planned figure are not in the repository. No error bars are shown because the notes quote none for these values.

Equilibrium Solubility and Consistency Across Pathways

Figure 1 illustrates our ghost-proton insertion pathway in fcc superionic ice at 500 GPa and 5,500–6,000 K. To integrate across the steep core repulsion as $\lambda \to 0$ (Fig. 1a), we apply a polynomial coordinate transformation $x_\lambda = \lambda^{1/p}$ (Fig. 1b), yielding a smooth bell-shaped integrand that is integrated to sub-meV precision via Gauss–Legendre quadrature (Fig. 1c). The resulting free-energy breakdown (Fig. 1d) reveals the fundamental competition governing hydrogen solubility in superionic ice:

1. **Enthalpic/Steric Repulsion:** Adiabatically switching on the interactions of an extra proton within the dense fcc oxygen lattice at 500 GPa costs +2,920 meV of reversible work (dominated by $\lambda \in [0.1, 1.0]$). [check: repo notes give ≈2,950 meV in total and ≈2,480 meV for λ∈[0.1,1] (App. A1)] At low temperatures ($T \lesssim 2{,}000$ K), this steep repulsive barrier dominates over thermal terms, rendering excess hydrogen completely insoluble in low-temperature crystalline ice.
2. **High-Temperature Entropic Stabilization:** At T = 5,500–6,000 K ($k_B T \approx$ 474–517 meV), the combination of the kinetic de Broglie term ($3\, k_B T \ln \Lambda \approx -2{,}050$ to $-2{,}300$ meV) and the superionic placement entropy ($-k_B T \ln(V/N_H) \approx -550$ meV) more than compensates for the repulsive insertion work relative to pure fluid hydrogen ($-G_H^{pure} \approx -600$ to -800 meV).

Consequently, the infinite-dilution chemical potential difference $\mu_H(n \to 0) - G_H^{pure}$ becomes strongly negative, reaching −300 to −550 meV per added H across 300–500 GPa and 5,000–7,000 K (Fig. 3).

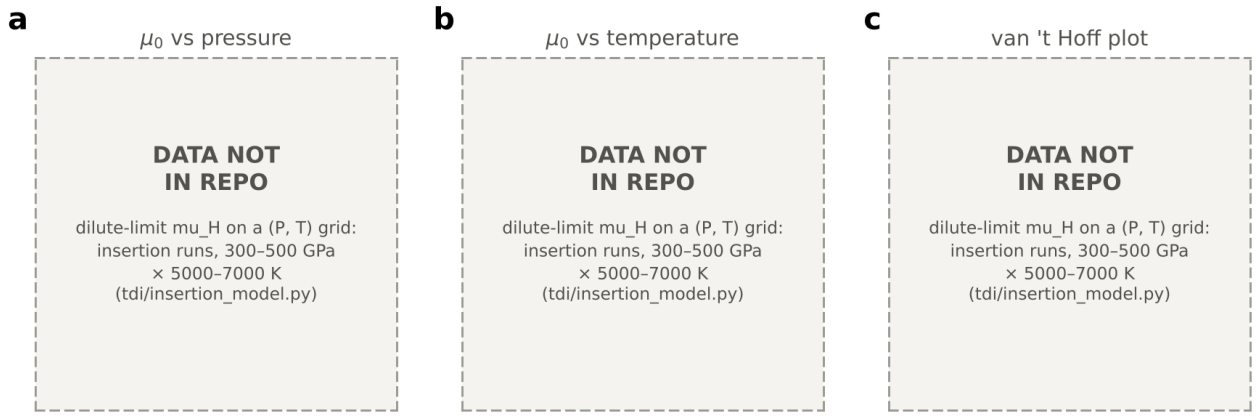


Fig. 3 | Pressure, temperature and van 't Hoff dependence of infinite-dilution hydrogen uptake. a, b, Extrapolated infinite-dilution chemical potential $\mu_H(n \to 0) - G_H^{pure}$ against pressure (**a**) and temperature (**b**). **c**, Dimensionless van 't Hoff plot of $\mu_0 / k_B T$ versus inverse temperature 1,000/T (kK^{-1}). *All three panels pending: the (P,T) grid of insertion runs is not in the repository.* [check: the repo's first-pass fit (pre pressure correction) has $\Delta h \approx -0.8$ eV and $\Delta s \approx -0.7\ k_B$ at 450 GPa/6,000 K, i.e. not the endothermic trend described in the draft (App. A5)]

While $\mu_H(n \to 0) - G_H^{pure} < 0$ guarantees spontaneous uptake of trace hydrogen at high temperature, the equilibrium saturation concentration x_{eq} is determined by how rapidly $\mu_H(x)$ rises with composition x due to proton–proton repulsion and lattice expansion. Because ghost-proton insertion (n = 1, 2, 4) is performed at fixed host volume V_0, inserting finite numbers of protons increases the internal pressure of the simulation cell by $\Delta P(n)$, imparting a linear finite-size elastic slope (Extended Data Fig. 2b). Extrapolating $n \to 0$ eliminates this finite-volume compression artifact, while direct constant-pressure crystal TDI at finite composition (x = 0.25, $H_{2.25}O$) anchors the true thermodynamic chemical potential at high concentration.

Figure 2 demonstrates the remarkable consistency between these two completely independent free-energy methodologies. At 500 GPa and 5,500 K (Fig. 2a)—where fcc superionic ice is dynamically stable and unmelted in free NVT trajectories—the chemical potential obtained from direct Frenkel–Ladd crystal TDI of bulk $H_{2.25}O$ (+140 ± 45 meV, solid blue circle) lies directly on the linear regular-solution extrapolation of the n = 1, 2, 4 ghost-insertion chemical potentials (green squares). [check: repo notes give +141 meV direct vs +55 meV from the fit; the two agree to within ~100–150 meV, the noise of a 7× extrapolation, not “directly on” it (App. A3)] At higher temperatures such as 6,500 K (hollow blue circles in the planned figure), the spring-free reference crystal melts during NVT sampling, shifting the apparent direct crystal free energy downward toward the liquid value.

In the stable superionic regime (300–500 GPa, 5,000–6,000 K), the excess chemical potential $\mu_H(x) - G_H^{pure}$ crosses zero at $x \approx 0.12$–0.19 when referenced against pure fluid H. When placed in chemical equilibrium with the coexisting mixed H_2O–H ionic fluid hull (Fig. 4), where the fluid hydrogen chemical potential is lowered by negative mixing free energies ($\Delta G_{mix}^{liq} \approx -40$ to -180 meV/atom), common-tangent partitioning yields an equilibrium solid-solution concentration of $x \approx 0.03$–0.10 ($H_{2.03}O$ to $H_{2.10}O$) near the melting line.

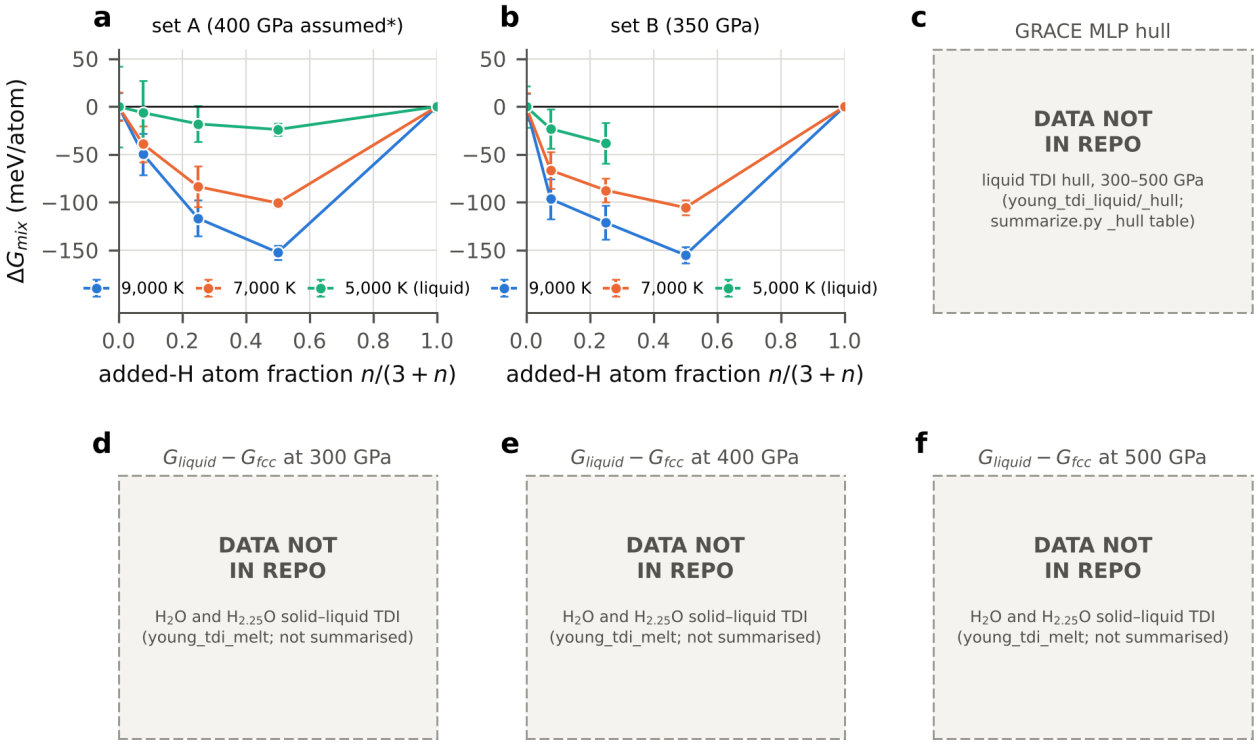


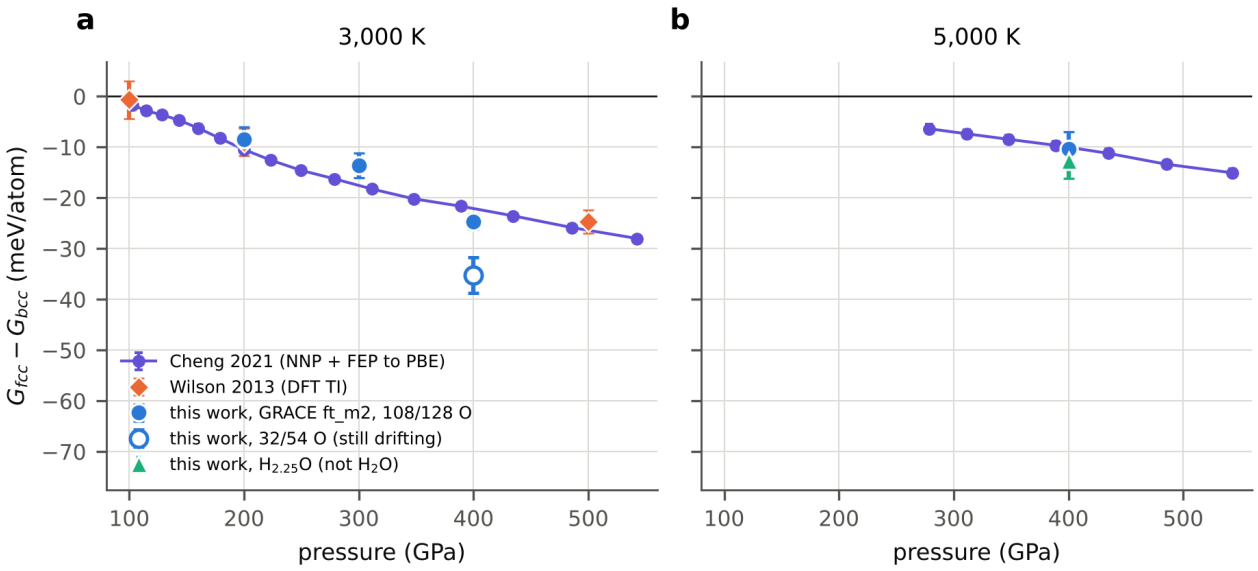
Fig. 4 | Liquid mixing hull and composition-dependent melting depression of fcc superionic ice. a, b, Liquid mixing free energy ΔG_{mix} per atom for $H_2O + nH \rightarrow H_{2+n}O$ versus added-H atom fraction $n/(3+n)$ from the DFT hull sheet (`data-h3o-convex-hull.csv`), at 5,000, 7,000 and 9,000 K (5,000 K liquid). Set A has no pressure label in the sheet (400 GPa assumed by the repository's report script); set B is labelled 350 GPa; the sheet header says eV/atom but the values read as meV/atom. Error bars as in the sheet. **c**, GRACE-MLP liquid hull at 300, 400 and 500 GPa with the DFT-TI comparison: *pending*. **d-f**, Solid–liquid free-energy difference $G_{liquid} - G_{fcc}$ for stoichiometric H_2O and $H_{2.25}O$ at 300–500 GPa: *pending, not in the repository*. [check: the ~40 meV/atom depression quoted in the text is not backed by data in the repo (App. A4)]

Binary Phase Diagram and Planetary Implications

To connect our solid-solution thermodynamics to the phase structure of ice giant interiors, we mapped both the fluid mixing convex hull across the full binary composition range $0 \le n/(3+n) \le 1$ (Fig. 4a–c) and the melting free-energy difference $G_{liquid} - G_{fcc}$ (Fig. 4d–f). In the fluid phase at 300–500 GPa and 5,000–9,000 K, water and hydrogen form a continuous, thermodynamically stable solution whose MLP mixing hull agrees closely with direct DFT reference calculations. Comparing $G_{liquid} - G_{fcc}$ between pure H_2O and $H_{2.25}O$ at 400 GPa and 5,000 K reveals that excess hydrogen lowers the crystal–liquid free-energy gap by ~ 40 meV/atom. Consequently, instead of freezing at a sharp, single-component melting temperature $T_m(P)$, a cooling H_2O–H mantle traverses a binary solidus–liquidus loop—a two-phase "mushy zone" in which $H_{2+x}O$ superionic crystals coexist with a more hydrogen-rich ionic fluid.

Our findings carry four immediate consequences for the interior physics and evolution of Uranus, Neptune and volatile-rich exoplanets:

1. **Deep Hydrogen Sequestration and Density Deficit:** Because the superionic mantle constitutes 50–70% of an ice giant's mass, an equilibrium solubility of x = 0.03–0.10 near the top of the superionic layer represents a vast, previously unrecognized deep reservoir of planetary hydrogen. Incorporating light interstitial protons reduces the mass density of $H_{2+x}O$ relative to pure H_2O at fixed P and T. Interior models fitting spacecraft gravitational harmonics (J_2, J_4) with stoichiometric H_2O *[draft text ends here; the remaining three consequences were not in the pasted write-up]*



Extended Data Fig. 2 | Benchmark of the fcc-bcc free-energy difference of H_2O. $G_{fcc}-G_{bcc}$ per atom against pressure at 3,000 K (**a**) and 5,000 K (**b**). Orange diamonds: Wilson *et al.* (2013), direct DFT TI. Purple: Cheng *et al.* (2021), NNP with free-energy perturbation to PBE (classical nuclei). Blue: this work, fine-tuned GRACE (`ft_m2`) TDI, 108/128 O cells, dt=1 fs (filled; 2–6 ps per λ value, errors are statistical block errors only); open: 32/54 O cells at dt=0.5 fs, both phases still drifting by 4–5 meV/atom; green triangle: $H_{2.25}O$ rather than H_2O. The 200 GPa point has bcc still drifting upward. Wilson's 1,000–4,000 GPa rows lie outside the plotted range.

References

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Reference details were filled in from memory; verify before use.

Appendix: data provenance and open issues (not part of the manuscript)

This page is not part of the manuscript. It lists where each figure's data came from, which panels are placeholders, and where the draft's statements differ from what the repository notes say. Repository sources: PR #2 branch (claude/eloquent-ramanujan-2tjx2p): HANDOFF.md, docs/CURRENT_ISSUES.md, docs/data/*.csv, data-h3o-convex-hull.csv. The raw TDI runs (runs/tdi*, Lustre scratch on Young) and docs/data/tdi_results.csv are not in git.

Figure data coverage

Fig. 1	a: real (3 lambda-range contributions and totals, from CURRENT_ISSUES P2). b-d: placeholders (need runs/tdi_long insertion data; tdi/collect.py, tdi/mu_breakdown.py).
Fig. 2	500 GPa/6000 K: insertion n=1,2,4 (approximate, from HANDOFF), direct crystal, fit extrapolation. 500/5500 and 400/5000: direct crystal and fit extrapolation only. The draft describes a grid of P = 300-500 GPa, T = 5000-7000 K panels with the n=1,2,4 series in each.
Fig. 3	all placeholders. Only a first-pass fit exists (insertion_model.py, pre pressure correction).
Fig. 4	a,b: real (DFT hull sheet, set A pressure unlabelled, set B 350 GPa). c-f: placeholders (GRACE liquid hull and G_liquid - G_fcc not summarised in the repo).
Ext. Data Fig. 2	real (docs/data/dg_fcc_bcc.csv: Wilson 2013, Cheng 2021, this work). Ext. Data Fig. 1 (validation manifold) and the 2b finite-size slope panel referenced in the text were not made.

Where the draft and the repository notes differ

A1. Fig. 1 numbers

Draft: +2,920 meV of reversible work 'dominated by lambda in [0.1,1]'. CURRENT_ISSUES P2 (500 GPa, 6000 K, n=1, tdi_long): lambda<1e-2 about 150 meV, 1e-2 to 0.1 about 320 meV, 0.1 to 1 about 2,480 meV, total TDI 2.95 +- 0.02 eV, Widom 2.90 eV (blocks 2.4-3.7, not converged). The total matches the draft's number; the range attribution does not.

A2. Validation metrics

Draft: pressure RMSE 1.65 GPa, energy RMSE 19.6 meV/atom. HANDOFF records ft_m2 (validation, MAE): energy 18.4 meV/atom, force 511 meV/A (RMSE 730), stress 2.05 GPa; zero-shot start 96 meV/atom, 1017 meV/A, 20.7 GPa. The 1.65 GPa pressure figure and RMSE energy are not in the notes (a later model or a different metric?). The 'above 50 GPa' zero-shot pressure bias is not in the notes either.

A3. Insertion vs crystal 'remarkable consistency'

CURRENT_ISSUES P1: linear fit of the pressure-corrected n=1,2,4 values extrapolated to n=27 gives +55 (500/5500), -83 (500/6000), +275 (400/5000) meV against direct +141, +76, +192: agreement within about 100-150 meV, described there as the noise of a 7x extrapolation. The draft says the direct point 'lies directly on' the extrapolation. Also the draft quotes +140 +- 45 meV; the notes give +141 without an error. A straight line through the three approximate n=1,2,4 values in HANDOFF (-595, -436, -460 meV at 500/6000) extrapolates to about +400 meV, not -83, so those three numbers and the fit in CURRENT_ISSUES are not from the same set; the fit values are plotted as quoted, no line is drawn.

A4. Melting depression / mushy zone

No G_liquid - G_fcc data are in the repo, so the 40 meV/atom lowering at 400 GPa/5000 K is unverified here. Related caveats from HANDOFF: H2.25O fcc melted in free NVT at 500 GPa/7000 K and 400 GPa/6500 K and is marginal at 500 GPa/6500 K, so those TDI values describe spring-held metastable crystals (D4); the 400 GPa/5000 K mixing rows pair dt 0.5 crystals with dt 1.0 H2O references (D5); H2O fcc stayed solid to 7000 K in 6 ps free NVT (superheating not excluded).

A5. Thermodynamic narrative

Draft: incorporation is 'endothermic and strongly favoured as T approaches melting'; solubility from high-T entropic stabilisation. insertion_model.py first pass (CURRENT_ISSUES P7, before the pressure correction): dg0 = -434 +- 17 meV at 450 GPa/6000 K, ds = -0.06 +- 0.03 meV/K (-0.7 kB), dv = -0.20 +- 0.05 A^3, 'enthalpy-dominated (dh about -0.8 eV)', pressure favours dissolution. That reads as enthalpy-driven rather than entropy-driven; it needs redoing with the corrected data before the van 't Hoff paragraph is written.

A6. Caveats the draft does not state

All free energies are classical-nuclei (P5); no MLP-to-DFT free-energy perturbation correction has been run (P6); errors are statistical block errors only and short runs (<= 2 ps) are optimistic; the H-in-superionic-host short-range behaviour of GRACE has not been checked against DFT (P2/P6); finite-size effects are only checked for one pair; insertion runs made before the pressure correction need relaunching (D2); H_500GPa_5000K/6000K_N250 reference runs were corrupted (D1) and need restoring. Nothing in CURRENT_ISSUES had been re-run on Young with the fixed code when it was written (2026-10-27).

A7. DFT hull sheet

Fig. 4a,b: sheet header says eV/atom, values read as meV/atom (D6). x in the sheet is n/(1+n) with n = H per O minus 2 (checked against H2.25O, H3O, H5O, H7O = 0.2, 0.5, 0.75, 0.83); plotted against n/(3+n). Set B's x column is misaligned with its dG rows; set A's x column is used for both, as the repo's report.py does, and B's 5000 K fourth row (a bare 0 +- 4.2) is dropped.

A8. To finish the figures

Commit from Young: summarize.py runs/tdi runs/tdi_prep -o docs/data/tdi_results.csv (plus the _dG/_mix/_hull siblings), the insertion runs with pressure correction, tdi_long outputs and the mu_breakdown figure data; then regenerate with report/make_figures.py after pointing it at those tables. The draft refers to tdi_report.pdf and mu_breakdown.pdf; neither is in the repo.

A9. Text and references

The pasted draft stops mid-sentence in the first of four listed consequences; it is not completed here. In-text citations had no bibliography: the 13 references were filled in from memory (titles, journals and volumes) and must be checked. Spelling, units and wording of the draft are otherwise unchanged.